Pyranometer



Category: **Solar, Power & Energy** Type: **Short SOP** Priority: **Medium** 4 labs

Covers these lab items: Pyranometer; Solar power meter (pyranometer)

What it is used for

To measure solar irradiance using a pyranometer. To obtain reliable readings by positioning the sensor correctly and following the instrument's measurement procedure.

Main parts

- Radiation sensor
- Protective dome or sensor surface
- Display or data logger
- Mounting base
- Signal cable or power connection

Before you start

1. Inspect the sensor surface, cable, and display for damage.
2. Check calibration status and operating instructions.
3. Place the sensor in an unobstructed location with the required sky view.
4. Ensure the sensor surface is clean and dry.
5. Confirm the correct units and data logging settings.

How to use

1. Mount the pyranometer level using its levelling feature where provided.
2. Connect the sensor to the approved display or data logger.
3. Switch on the system and allow the required stabilization time.
4. Check the sensor orientation and confirm that shadows do not fall on it.
5. Record irradiance readings under the required conditions.
6. Note the time, weather conditions, and measurement location.
7. Transfer or save the readings according to the procedure.
8. Clean the sensor surface and secure the instrument after use.

⚠ Safety

- Follow electrical safety instructions for powered data loggers.
- Use suitable precautions when working outdoors or at height.
- Do not touch hot or energized electrical connections.
- Avoid looking directly at the sun while positioning the instrument.

Care & storage

- Clean the sensor surface using the manufacturer's approved method.
- Inspect cables, seals, and mounting hardware regularly.
- Keep the instrument level and calibration current.
- Store it in a protective case when not installed.

Common mistakes to avoid

- Allowing shadows to fall on the sensor.
- Using an unlevel or dirty sensor.
- Ignoring calibration status or units.
- Recording readings without noting time and conditions.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.

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